PRODUCT DATA SHEET

DUROFLOW 420

DUROTECH
Engineered Construction Chemicals
Made In Australia For Over 45 Years

DESCRIPTION

Duroflow 420 is a cementitious, non-structural, bulk fill floor underlayment with exceptional adhesion properties. Duroflow 420 is suitable for internal applications where a thickness of between 5mm and 80mm is required. Use Duroflow 420 over new or old concrete floors prior to the application of floor coverings. Can be used for bulk filling applications, creating ramps and falls in screeds.

FEATURES

- May be applied to a thickness of up to 80mm in one application.
- Fast Setting
- Tear and rain resistant PE bags which are recyclable and reduce product loss from damaged packaging
- Highly accurate and consistent bag weights
- Excellent adhesion
- Exceptional workability
- Can also be screeded to create floor grades, ramps or falls to waste
- Trafficable at 2 hours
- Accepting of certain floor coverings after 4 hours – refer to Temperature Considerations

USE AREAS

- Creating floor grades, ramps or falls to floor wastes
- Indoor applications
- Residential and commercial applications
- New construction
- Refurbishment of old floors
- Underlayment for carpet, carpet tiles and ceramic tiles after 4 hours cure – refer to Temperature Considerations
- Underlayment for vinyl and timber after 24 hours cure – refer to Temperature Considerations

Surface Preparation

Concrete floors must be structurally sound, clean and dry. Surface must be free from dust, dirt, wax, grease, asphalt, latex and gypsum compounds, adhesives, paint, curing and sealing compounds and other contaminants

which may act as a bond breaker. Concrete must be free from laitance, efflorescence and not be subject to hydrostatic pressure. Mechanically prepare the floor using recommended preparation methods such as shot blasting, scarifying, diamond grinding, or other suitable methods to provide a roughened, clean, sound, solid and open porous surface. Acid etching is not a suitable method of preparing the subfloor. Remove all dust and debris from the floor by vacuuming the surface with a suitable H Class industrial cleaner – must be equipped with a Hepa filter.

All concrete sub floors must be fully cured and dry in accordance with AS 1884 (Less than 75% relative humidity

when measured in accordance with ASTM F2170). Ambient temperature, surfaces and materials should be maintained at temperatures higher than 10°C and below 35°C. For floors with a high humidity content or subject .

All concrete sub floors must be fully cured and dry in accordance with AS 1884 (Less than 75% relative humidity when measured in accordance with ASTM F2170). Ambient temperature, surfaces and materials should be maintained at temperatures higher than 10°C and below 35°C. For floors with a high humidity content or subject to rising damp, apply Durotech Moisture Vapour Barrier. Refer to the Durotech Moisture Vapour Barrier TDS for details. Refer to Durotech Technical Bulletin on Concrete Subfloor Preparation for detailed information on preparation of the subfloor.

SINCE 1977

Priming

Using a soft brush or broom, prime the prepared concrete area with Durotech Primer and allow primer to dry.

A second coat of primer may be necessary on areas with very porous surfaces where the initial coat has been completely absorbed. Allow the primer to fully dry to a clean, thin film (approx. 1- 2 hours depending on ambient conditions) before applying Duroflow 420. Low temperature can delay the drying time of primer. Do not apply levelling until the primer has dried thoroughly.

Mixing

Place the recommended amount of potable water into a clean mixing vessel and whilst mixing, slowly add the entire 20kg contents of Duroflow 420. Mixing should take place using a forced action high shear mixing paddle, with a mixer capable of stirring at 600rpm. Mix for approximately 2 – 3 minutes to a lump free consistency. Use mix within 15 minutes after mixing, be sure to only mix a quantity that can be use within this time.

CONSISTENCY	LITRES / 20KG BAG
Levelling application	**2.7 – 2.9**
Screed application	**2.2 – 2.3**

- Do NOT overwater the Duroflow 420, use only the recommended mix water volume.
- Do NOT use a concrete or masonry mixer to mix Duroflow 420.
- Do NOT mix by hand.
- Do NOT attempt to retemper mixed product once the working time of 30 minutes has been exceeded.

Application:

Levelling Application

Pour the mixed Duroflow 420 onto the prepared substrate and spread into place using a long handled gauged spreader. Duroflow 420 seeks its own level during the first 10 - 15 minutes after pouring. Subsequent applications of leveller should be made whilst the material on the floor is still fluid and has a wet edge, to allow for easy blending. It is recommended that spiked boots be worn, to minimise any indentations in the product whilst still fluid.

Screed Application

Pour the mixed Duroflow 420 onto the prepared substrate and spread into place using a screed bar or power screed to the desired thickness or fall. The Duroflow 420 can be applied up to 80mm thick in one application. Thicker applications may require a longer cure.

Set Time

Allow a cure time of 2 hours at 23°C for foot traffic. Allow the Duroflow 420 to cure 4 hours at 23°C before applying carpet, carpet tile of ceramic tile floor coverings. Allow the Duroflow 420 to cure 24 hours at 23°C before applying impervious floor coverings, such as timber or vinyl. These times can be impacted by air temperature, substrate temperature and relative humidity. Thicker layers will extend drying times.

Coverage

20kg bag of Duroflow 420 will cover approximately 1.1m2 at 10mm thickness for a leveller consistency, and, 1.0m2 at 10mm thickness for a screed consistency.

Clean Up

Wash all tools in water immediately after use.

Temperature Consideration

The mechanism of interaction between cement and water is temperature sensitive. The set time is delayed at low temperatures and is accelerated at high temperatures. To avoid significant change in setting times, the recommended water temperature and ambient temperature ranges are:
Water Temperature Range: 15 – 25˚C. Working with temperatures outside of this range will also impact the fluidity of the grout.
Ambient Temperatures: Do not grout at a temperature less than 10˚C. Above 30˚C, consider using cooled water for mixing the product. Do not grout in temperatures above 35˚C.
Substrate Temperature: Do not apply onto a surface which has a temperature less than 10˚C or above 35˚C.

Precautions

- This product is designed as an underlayment and must not be used as a trafficable wearing surface.
- Do not use externally or in areas where the underlayment or subfloor is likely to be exposed to submersion, wetting or high levels of humidity.

Packaging

20kg Polyethylene (PE) bags.

Shelf Life & Storage

Shelf Life

The shelf life of the product is 18 months from the date of manufacture, if stored indoors in accordance with recommended storage conditions.

Storage:

Store in dry conditions, in unopened and undamaged PE bags and in temperatures below 30˚C. If stored in excessive temperature conditions, externally exposed to the elements or in high humidity conditions, the shelf life may be reduced.

Duroflow 420 – Product Data:

Property	Test Result - Levelling	Test Result - Screed
Working Time at 23˚C	15 minutes	10-15 minutes
Fresh Mix Density (kg/m³)	2015	2166
Yield (L/20kg Bag)	10.8L @ 2.7L water addition 10.9L @ 2.9L water addition	10.2L @ 2.2L water addition 10.3L @ 2.3L water addition
20 Kg Bags per cubic metre	93 @ 2.7L water addition 92 @ 2.9L water addition	98 @ 2.2L water addition 97L @ 2.3L water addition
Foot Traffic at 23˚C	Approx. 2 hours	Approx. 2 hours
Floor Coverings at 23˚C	Approx. 2 hours	Approx. 2 hours
Setting Times at 23 ˚C	Initial: 45 minutes Final: 60 minutes	Initial: 20 minutes Final: 40 minutes
Compressive Strength	1 Day: 17 - 18 MPa 28 Days: 32 - 33 MPa	1 Day: 39 - 40 MPa 28 Days: 54 - 55 MPa
Application Thickness	Minimum: 5mm Maximum: 80mm	Minimum: 5mm Maximum: 80mm

Testing Parameters: 14% of water for levelling consistency 11.25% for screed consistency. Laboratory at: 23±2˚C > 50% RH.
The performance data is typical and based upon controlled laboratory conditions. Actual performance on the job site may
vary from these values based on actual site conditions .

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